

IoT Hacking: Attacking the Photon

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ABSTRACT

The Internet of Things (IoT) is becoming increasingly prevalent as the world moves towards the vision of a smarter home and city. As more of these products, which range from interconnected sensors to wearable devices, get connected to the Internet, serious concerns are being raised with regards to its security. Unfortunately, as the domain of IoT is relatively new, security is inadequately considered in these devices. This paper explores the security aspects of a popular IoT device, the Particle Photon. Furthermore, the paper demonstrates a viable attack to compromise a Particle Photon, and retrieve its stored Wi-Fi passwords over the Internet. Finally, the paper proposes potential defense measures that can be adopted to better secure the Particle Photon.

Categories and Subject Descriptors

C.2.1 [Computer-Comm. Networks]: Network Architecture and Design; C.2.3 [Computer-Comm. Networks]: Network Operations; C.2.6 [Computer-Comm. Networks]: Internetworking; C.3 [Special-purpose and application based systems]

General Terms

Security.

Keywords

IoT, Particle, Photon.

1. INTRODUCTION

Internet of Things (IoT) refers to the interconnected network of physical objects that enables the collection and exchange of data over the Internet. Examples include using IoT to monitor and manage energy consumption within a household, remotely switching on/off appliances, and automatically unlocking of doors. Contributing to the revolution of smart cities worldwide, advanced Information and Communication Technologies (ICTs) infrastructures have implemented IoT as one of the key technologies supporting this growth.^[6] This shows the importance of IoT in today's market and how it will be widely used in future. Furthermore, with this increased involvement of IoT on critical data over the Internet, questions on the security of IoT devices should be raised and investigated. In 2014, a researcher in Proofpoint uncovered a botnet where 25% of the traffic comprised of compromised IoT devices^[5]. This incident demonstrated the

potential risk of unsecured IoT devices, and further affirms the importance of IoT security.

Out of all of the IoT devices in the market, the Particle Photon was chosen because the device is a popular IoT development kit and is relatively inexpensive, at US\$19. The Particle Photon, henceforth Photon, is a Wi-Fi enabled development kit meant for IoT prototyping. The Photon is a product by Particle, and is a part of a family of development kits, including kits with cellular connectivity. The Photon leverages on another product by Particle, a P0/P1 chip, which contains a STM32F205 120Mhz ARM Cortex M3 microcontroller and a Broadcom BCM43362 Wi-Fi chip that communicates on 802.11b/g/n protocol. The Photon also comes with a 1MB flash memory and 128KB RAM which are preloaded with Particle's firmware libraries. Because Particle products are meant for prototyping and subsequent deployment, the Photon has both open sourced hardware and software.

In this paper, the security aspects of the Photon will be explored. Additionally, a viable attack to compromise a Particle Photon and retrieve its stored Wi-Fi passwords over the Internet will be demonstrated. Finally, the paper will propose potential defense measures that can be adopted to better secure the Particle Photon.

2. BACKGROUND

2.1 Particle Platform

Before the Photon's features can be accessed, a user has to sign up for a Particle account. Upon signup, the account is entitled access to the Particle Platform, which include tools such as the web IDE, mobile SDK, and the device dashboard.

2.2 Access Token and Device ID

Each particle account is assigned an access token, a 40 digit hexadecimal string that is used to interact with the account and its associated devices through the use of Particle's Cloud API.^[3] By default, the assigned access tokens do not expire, but users are allowed to assign an expiry date to them.